

Once-Daily Valbenazine for Tardive Dyskinesia or Huntington's Disease Chorea in Patients With Dysphagia or Pill Aversion

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BACKGROUND

- Valbenazine is a uniquely selective, once-daily medication approved for tardive dyskinesia (TD) and chorea associated with Huntington's disease (HD)¹
- For patients with TD and HD, swallowing whole medications may be difficult due to their motor symptoms and primary condition^{2,3}
- Pill aversion (or non-physiologic dysphagia) can arise from psychological or emotional factors, such as fear of choking in patients with HD chorea²
- Dysphagia may be particularly problematic in elderly patients, due to anatomical and physiologic changes associated with aging⁴
 - The presence of comorbid conditions (e.g., gastrointestinal reflux, Alzheimer's disease) and medication side effects (e.g., dry mouth) can further exacerbate problems with swallowing
- Alternative modes for medication administration are needed for patients with dysphagia or pill aversion
- Three in vitro studies assessed the acceptability of crushing the contents of valbenazine capsules⁵
 - Study 1: established the dissolution profile of crushed capsule contents
 - Study 2: tested compatibility of crushed capsule contents with soft foods and liquids
 - Study 3: evaluated the suitability of delivering crushed capsule contents via gastrostomy tube (G-tube)
- In addition, a new and recently approved valbenazine granules formulation was evaluated in 2 studies:
 - Study 4 (in vitro): tested compatibility of valbenazine granules with soft foods
 - Study 5 (phase I): established bioequivalence between valbenazine granules and capsules containing compacted powder plug and between fed and fasted states for granules

OBJECTIVE

- To evaluate the acceptability of various modes of administration for the 2 approved valbenazine formulations: capsule with compacted powder-plug content and sprinkle capsule with granules content

METHODS

- Methods for the 4 in vitro studies are summarized in **Table 1**
- The phase I trial was an open-label, 2-cohort study with healthy adults (**Table 2**)
 - Cohort 1 (bioequivalence cohort): 3-sequence, 3-period crossover design with 24 participants
 - Cohort 2 (food effect cohort): 2-sequence, 2-period crossover design with 14 participants
- Pharmacokinetic (PK) parameters for valbenazine and its primary metabolite, $[+]-\alpha$ -dihydrovalbenazine ($[+]-\alpha$ -HTBZ), were used to determine bioequivalence: maximum plasma concentration (C_{max}) and area under the curve from 0 hours to the last measurable concentration (AUC_{0-10hr}) and to infinity ($AUC_{0-\infty}$)
- Bioequivalence between valbenazine granules and compacted powder capsule (Cohort 1) and between fed and fasted states for granules sprinkled on applesauce (Cohort 2) was defined as the 90% CI for the geometric mean ratios of PK parameters falling between 80% to 125%

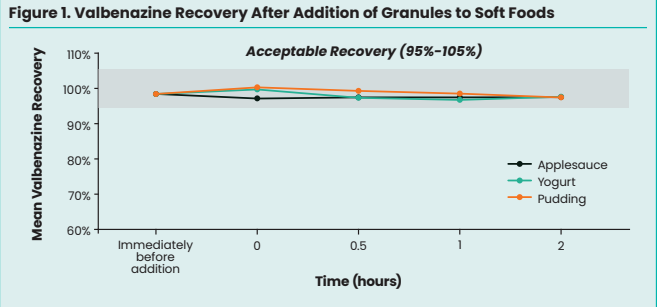
Table 1. Studies 1-4 (In Vitro)	
Study	Methods
With Crushed Contents of Valbenazine Capsules ^a	
Study 1: Dissolution Profile ⁵	- Samples were prepared for 2 doses (40 mg, 80 mg) - Whole, unopened capsules (reference) and crushed capsule contents (test) were added to an HCl dissolution bath - Samples were collected at 10, 15, 20, 30, 45, and 60 minutes and analyzed using HPLC
Study 2: Food Compatibility ⁵	- Crushed capsule contents (40 mg, 80 mg) were mixed with 1 Tbsp of applesauce, yogurt, or pudding - Crushed contents (40 mg) were also added to buffer solutions (pH 1.2, 4.5, and 6.8) and FeSSGF - Samples were analyzed from immediately after addition of crushed contents (0 hours) to 2 hours post-addition - Acceptability (i.e., stability in tested foods/liquids) was defined as 90%-110% valbenazine recovery
Study 3: G-tube Suitability ⁵	- Crushed capsule contents were mixed into 10 mL of water and added to a G-tube - Various water temperatures, G-tube diameters, and storage temperatures were evaluated, along with the need for a cup rinse - Acceptability (i.e., suitability for G-tube administration) was defined as 90%-110% valbenazine recovery
With Valbenazine Granules (New Approved Formulation)	
Study 4: Food Compatibility	- Uncrushed granules were mixed with 1 Tbsp of applesauce, yogurt, or pudding - Granules were also added to buffer solutions (pH 1.2, 4.5, and 6.8) and FeSSGF - Samples were analyzed from immediately after addition of granules (0 hours) to 2 hours post-addition - Acceptability (i.e., stability in tested foods/liquids) was defined as 95%-105% valbenazine recovery

^aCapsules contain a plug of compacted powder.
FeSSGF, fed-state simulated gastric fluid; G-tube, gastrostomy tube; HCl, hydrochloric acid; HPLC, high-performance liquid chromatography; Tbsp, tablespoon.

Table 2. Study 5 (Phase 1)				
Cohort 1: Bioequivalence Cohort (N=24)				
Treatment Period	1		2	3
Treatment Sequence	1	Granules sprinkled on applesauce	Whole capsule with granules	Whole capsule with compacted powder
	2	Whole capsule with granules	Whole capsule with compacted powder	Granules sprinkled on applesauce
	3	Whole capsule with compacted powder	Granules sprinkled on applesauce	Whole capsule with granules
Cohort 2: Food Effect Cohort (N=14)				
Treatment Period	1		2	
Treatment Sequence	1	Granules sprinkled on applesauce plus meal (fed state)	Granules sprinkled on applesauce (fasted state)	
	2	Granules sprinkled on applesauce (fasted state)	Granules sprinkled on applesauce plus meal (fed state)	

RESULTS

- Studies 1-3 with the crushed contents of valbenazine capsules demonstrated the following:
 - Study 1: dissolution profile of crushed capsule contents was similar to whole intact capsules; rapid and complete valbenazine release was observed regardless of preparation (crushed contents or whole capsule) and dose (40 mg or 80 mg)
 - Study 2: valbenazine recovery was acceptable within 2 hours after capsule contents were added to soft foods, buffer solutions, and fed-state simulated gastric fluid (FeSSGF); on the basis of pH and common use, other acceptable foods and liquids include: cottage cheese, hummus, jelly/jam, mashed fruits, mashed potatoes, and orange and cranberry juice
 - Study 3: G-tube administration was suitable when crushed valbenazine contents were added to room-temperature water and stored in water for up to 2.5 hours
- Study 4 showed valbenazine granules to be compatible with applesauce, yogurt, and pudding (**Figure 1**)
 - Valbenazine recovery was also acceptable for buffer solutions (pH 1.2 and pH 4.5) and FeSSGF, indicating compatibility with a wide range of acidic to slightly acidic soft foods (e.g., cottage cheese, hummus, jelly/jam, mashed fruits, and mashed potatoes)



- Study 5 indicated bioequivalence between valbenazine granules and the compacted powder formulation (Cohort 1) and between fed and fasted states for valbenazine granules when sprinkled on applesauce (Cohort 2) (**Table 3**)
 - In Cohort 1, there was a slight delay in absorption rate (valbenazine C_{max}) but no effect on overall absorption (valbenazine AUC, $[+]-\alpha$ -HTBZ AUC)

Table 3. Bioequivalence Assessments for Valbenazine Granules				
Analyte	Parameter	Geometric Mean Ratio (90% CI)		
		Granules Whole Capsule ^a	Granules on Applesauce ^a	Granules on Applesauce + High-Fat Meal ^b
Valbenazine	AUC _{0-10hr} (ng·hr/mL)	93.2 (88.9, 97.7)	90.3 (85.7, 95.2)	102.2 (95.7, 109.2)
	AUC _{0-10hr} (ng·hr/mL)	93.2 (88.9, 97.7)	90.2 (85.6, 95.0)	102.1 (95.7, 108.9)
	C _{max} (ng/mL)	92.0 (80.9, 104.7)	71.8 (63.5, 81.1)	85.5 (76.0, 96.3)
$[+]-\alpha$ -HTBZ	AUC _{0-10hr} (ng·hr/mL)	89.9 (82.2, 98.4)	88.7 (81.9, 95.9)	103.6 (97.0, 110.7)
	AUC _{0-10hr} (ng·hr/mL)	89.8 (81.9, 98.4)	88.2 (81.7, 95.2)	103.8 (97.3, 110.7)
	C _{max} (ng/mL)	90.1 (84.7, 95.9)	90.6 (85.9, 95.6)	97.5 (87.6, 108.5)

^aversus whole capsule with compacted powder plug (Cohort 1)

^bversus fasted state (Cohort 2)
^c $[+]-\alpha$ -HTBZ, $[+]-\alpha$ -dihydrovalbenazine; AUC, area under the curve; AUC_{0-10hr}, AUC from 0 hours extrapolated to infinity; AUC_{0-10hr}, AUC from 0 hours to the last measurable concentration; C_{max}, maximum plasma concentration.

MODES OF VALBENZAZINE ADMINISTRATION

Compacted Powder Formulation^{1,2}

Swallow whole capsule with water

- Take pill as directed with water, with or without food

Crush and mix with soft foods

- Open capsule, place contents on spoon, and crush with back of second spoon
- Add crushed contents to ~1 Tbsp of soft food and mix
- Consume immediately or store for ≤2 hours

Crush and stir into drinks

- Open capsule, place contents on spoon, and crush with back of second spoon
- Add crushed contents to water or juice and stir
- Consume immediately or store for ≤2 hours

Crush, disperse in water, and administer via feeding tube

- Open capsule, place contents on spoon, and crush with back of second spoon
- Add crushed contents to ~10 mL of water and stir
- Draw mixture into a syringe and administer into feeding port immediately or store for ≤2 hours

Granules Formulation (NEW)¹

Swallow whole capsule with water

- Take pill as directed with water, with or without food

Mix with soft foods

- Open capsule, sprinkle granules onto ~1 Tbsp of soft food, and stir
 - Do not crush granules while stirring
- Consume immediately or store for ≤2 hours



Soft foods may include:

- Applesauce
- Yogurt
- Pudding
- Cottage cheese
- Hummus
- Jelly or jam
- Mashed fruits
- Mashed potatoes

CONCLUSIONS

- Results from 3 in vitro studies indicate the acceptability of crushing the compacted powder contents of valbenazine capsules for mixing with soft foods/liquids or dispersing in water for delivery via G-tube
- Results from an in vitro study of the new valbenazine granules formulation indicate compatibility with a wide range of acidic to slightly acidic soft foods (below pH 6.8)
- A phase I study demonstrated the bioequivalence between valbenazine formulations (granules vs compacted powder), with no food effects for valbenazine granules (fed vs fasted state)
- To date, valbenazine is the only VMAT2 inhibitor approved for TD and HD that offers a once-daily formulation with multiple routes of administration
- With 2 approved and available formulations, valbenazine offers alternative modes of administration that are suitable for patients with swallowing difficulties, including dysphagia and pill aversion

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Disclosures: This study was supported by Neurocrine Biosciences, Inc. (San Diego, CA). Writing and graphical assistance were provided by Citrus Health Group, Inc. (Chicago, IL). Please contact medinfo@neurocrine.com with any questions.

PRESENTED AT THE NATIONAL ASSOCIATION
OF SPECIALTY PHARMACY ANNUAL MEETING
SEPTEMBER 14-17, 2025; DENVER, CO

